

Astro Quiz Question Bank — Coverage Report

300 questions (ids 21–320) analysed against the 253-concept astronomy graph · quiz format: 20 questions in 180 seconds · July 2026

1 · Headline

90 / 253

CONCEPTS TESTED (36%)

142QUESTIONS ON JUST 24
CONCEPTS**55**QUESTIONS MATCHING NO
CONCEPT**289 / 11 / 0**

RECALL / APPLY / REASON

43%OF STUDY-PDF CONCEPTS
TESTED**~44**

QUESTIONS NEEDED REPAIR

The shape of the bank: concentrated, not broad. Half of all concept-matched questions pile onto 24 concepts (moons ×13, comets ×12, ISRO launch dates ×9, zodiac ×8...), while two-thirds of the concept graph goes untested. 78% of questions land inside study-material territory — but they re-ask the same patch of it. And the bank tests memory almost exclusively: 289 of 300 questions are pure recall; not one requires chained reasoning.

What is already right: question length. Median 17 words including all four options, max 33 — comfortably inside the 9-second-per-question budget. Expansion needs no format change, only content.

2 · Coverage by strand

STRAND	CONCEPTS TESTED		VERDICT
Stars, Galaxies & Cosmos	33 / 50	66%	strong
Solar System	29 / 45	64%	strong
Naked-Eye Sky	10 / 30	33%	weak — the verifiable-outdoors strand
Telescopes & Observing	9 / 30	30%	facility trivia, not optics
Light & Optics	3 / 10	30%	weak
Celestial Sphere & Motion	3 / 11	27%	weak — feeds the practical round
Time & Calendars	2 / 11	18%	near-absent
Indian Astronomy	1 / 21	5%	one question — in Aryabhat's own quiz
Astrophotography	0 / 24	0%	absent
Science & Geography Basics	0 / 21	0%	absent (defensible — prerequisites)

3 · Similar vs diverse — the concentration picture

GROUP	QUESTIONS	WHAT IT MEANS
Similar coverage — clustered on 24 concepts	142 (47%)	moons-of-planets ×13, comets ×12, India launch dates ×9, zodiac ×8, Sun-as-star ×8, solar activity ×7, observatory sites ×7...
Diverse — sole question on their concept	36 (12%)	the genuinely broad tail
Moderate — 2–3 questions per concept	67 (22%)	healthy
Unmappable trivia	55 (18%)	discovery months, facility names, mission pilots — no syllabus covers them, so no student can prepare

4 · Defects found — and repaired

The repair pass produced **astroquiz_question_repaired.csv** (original untouched). 300 in, 300 out.

CLASS	COUNT	ACTION TAKEN	EXAMPLES
Wrong answer	4	corrected	#123 true answer (two moons > Mercury) was not among options; #312 Milky Way marked eight-armed; #33 "Singularity" as the theory of spacetime; #45 zodiac width
Outdated	7	updated or reframed	the largest-solar-telescope cluster answered for two different eras — the bank contradicted itself (#212/282–284 vs #247, now unified on Inouye 4 m, Maui); "which two planets have rings" premise; farthest-object = quasar
Ambiguous	15	10 sharpened, 4 replaced, 1 kept	"magnetic cycle" keyed as 11 y (that is the sunspot cycle — stem fixed); 12 signs vs 13 ecliptic constellations; "first animal in space" vs in orbit
Duplicates	13	worse twin replaced	"How old is our Sun?" verbatim twice with the answer at different positions (#179/#301); Taurus-is-a-bull twice; Newton-and-mirrors twice
Mangled text	6 rows	fixed	° printed as º; Greek letters corrupted in star names (#107)

The 12 replacement questions target the gaps

Every replacement lands on an empty or starved strand, stays inside the length envelope, and several are apply-level: *Who wrote the Aryabhatiya* · *equinox sunrise direction* · *a star rises 4 min earlier each night* · *Polaris from the equator* · *phase after full moon* · *Chandrayaan-3 landing site* · *Jantar Mantar* · *aperture vs magnification (×2)* · *star-trail photos* · *festival calendars* · *synodic month*.

Structural tell left for the editorial pass: "None of these" appears as an option 25 times and is **never** the correct answer. A coached student eliminates it for free. Fixing it means per-question rewrites — queued for the expansion waves, not a mechanical pass.

5 · Expansion plan — 300 → 600 questions

STRAND	ADD	CONTENT
Naked-Eye Sky	+45	phases, motions, seasonal skies, planet-spotting — answerable by looking up
Indian Astronomy	+40	Aryabhata, Bhaskara, jantar mantars, panchang, nakshatras, ISRO <i>science</i>
Solar System (apply-level)	+35	why Venus is hottest, eclipse geometry, tides, seasons on Mars
Stars & Cosmos (apply-level)	+40	red = cool, lifetime vs mass, distance ladder reasoning
Celestial Sphere & Motion	+30	coordinates, circumpolarity by latitude, sidereal vs solar
Telescopes & Observing	+30	optics over facilities: aperture, focal ratio, mounts
Time & Calendars	+25	leap rules, time zones vs solar time, month origins
Light & Optics	+20	spectra, redshift, twinkling, atmospheric windows
Astrophotography	+20	exposure, tracking, trails, phone-at-eyepiece
Science Basics	+15	gravity, density, states of matter — the easy floor

Rules for every new question

- **Concept-tagged** — coverage stays a query, not an audit. The 55 unmappable questions are the cautionary tale.
- **Preparable** — answerable from the study material, not from a 1997 almanac.
- **Level mix** ~50/40/10 recall/apply/reason (bank today: 96/4/0). Apply fits 9 seconds: *"Full moon rises at? — sunset / midnight / sunrise / noon."*
- **Envelope**: stem ≤ 12 words, options ≤ 4 words.
- **Paper assembly by strata**: each 20-question paper draws a fixed profile (e.g. 8 recall + 8 apply + 4 hard) so rounds are comparable.
- **Authored in the concept-tagged quiz layer, exported to CSV** — validation, machine-balanced answer positions, misconception-bearing distractors, and "which PDF concepts still lack questions" becomes a CLI query.

Sources: astroquiz_question.csv (300 questions, ids 21–320; ids 1–20 are demonstration questions, excluded) · concept graph concepts/data/ (253 astronomy concepts of 562 total) · per-question tags concepts/coverage/astroquiz-bank-tags.json · repair log concepts/coverage/astroquiz-bank-repairs.md · repaired bank ~/Downloads/astroquiz_question_repaired.csv